JUNXIANG YANG

50 Nanyang Avenue, Singapore, 639798
(65) · 85906263 ◇ (86) · 13825354208 ◇ junxiang.yang@ntu.edu.sg
Google Scholar ◇ ORCID: 0009-0005-2036-9694 ◇ LinkedIn: junxiang-yang-047b80283

EDUCATION

Nanyang Technological University *Jan. 2025 – Present*
Ph.D. in Electrical and Electronic Engineering GPA: 4.17/5.0

University of Science and Technology of China *Sept. 2020 – June 2024*
B.S. in Physics Major: Applied Physics GPA: 3.21/4.3
Selected Courses: Electronic Circuits (93), Principle of Microcomputer (90)

Nanyang Technological University *July 2023 – Oct. 2023, Feb. 2024 – Apr. 2024*
Research internship in School of Electrical and Electronic Engineering

EXPERIENCES

Ph.D. Student / Research Assistant *Sept. 2024 – Present*
Advisor: Prof. Yun Yang *NTU EEE, Singapore*

- Designed **PCB-based electromagnetic metasurfaces** for MHz wireless power transfer and validated their voltage enhancement through simulation and experiments.
- Developed **two-stage FEM-assisted optimization strategies** for compact PCB-based resonant couplers in capacitive and hybrid wireless power transfer systems.
- Optimized **sandwich-structured PCB resonators** to improve equivalent quality factor, system efficiency, and misalignment tolerance.
- Built **data-driven prediction models** for equivalent-circuit electrical parameters of PCB-based couplers to accelerate design iteration.

Development of Host Software for X-ray Photoelectron Spectroscopy *May 2024 – Apr. 2025*
Advisor: Prof. Changqing Feng *USTC, Hefei*

- Designed a **user interface** using PyQt5, providing an intuitive GUI for experimental operation.
- Developed and tested a communication module for reliable information exchange with FPGA and detector.

PCB-Based Wireless Power Resonator Research Internship *July 2023 – Oct. 2023, Feb. 2024 – May 2024*
Advisor: Prof. Yun Yang *NTU, Singapore*

- Simulated polygonal planar spiral coils in HFSS and validated the design with PCB prototypes.
- Designed and optimized **13.56 MHz PCB resonators** with high quality factor for wireless power transfer.
- Developed an automated optimization workflow combining machine learning and finite element simulation.

Design of High-Voltage Circuits for MeV Gamma-Ray Detector *July 2022 – May 2023*
Advisor: Prof. Changqing Feng *USTC, Hefei*

- Designed the schematic and PCB of the power module for a MeV gamma-ray detector.
- Validated the power module and mitigated noise issues.

TECHNICAL STRENGTHS

Computer Languages	Python, MATLAB
Simulation Software	Ansys HFSS, Ansys Maxwell, COMSOL Multiphysics
English	TOEFL (87), IELTS (7.5), GRE (319+3.5)

HONORS AND AWARDS

First Place, Student Design Competition, IEEE Wireless Power Technology Conference and Expo (WPTCE), Halifax, Nova Scotia, Canada, Jul. 2026
Best Paper Award, 20th IEEE Conference on Industrial Electronics & Applications (ICIEA, Aug. 2025)
Finalist Certificate, 2024 IEEE Global Student Wireless Power Competition (SWPC, May 2024)
Certificate of Merit & **Certificate of Third Prize**, The Student Competition on Electric Transportation Design – Automotive, Vessel and Aircraft (May 2024)

LEADERSHIP

President of the Student Union, School of Physical Science *Apr. 2022 – Apr. 2023*

PUBLICATIONS

- **Junxiang Yang**, Xiangrong Zhang, Yici Wang, Shuye Shang, Kaiyuan Wang, Yun Yang, "A Frequency-Adjustable PCB Shielding Coil in Wireless Power Transfer System," in *Wireless Power Transfer*, Jul. 2024.
- Shuye Shang, Kaiyuan Wang, Rui Liang, Ke Tang, **Junxiang Yang**, Yun Yang, "Investigations on the Use of a Wireless Power Modulator in a Wireless Charging Container," in *Wireless Power Transfer*, vol. 11, e010, 2024, doi: 10.48130/wpt-0024-0010.
- **Junxiang Yang**, Kaiyuan Wang, Jiayang Wu, Yun Yang, "Two-Stage Optimization of DC Equivalent Series Resistances in Compact PCB-Based Resonant Couplers for Capacitive Power Transfer," in *20th IEEE Conference on Industrial Electronics and Applications (ICIEA)*, Yantai, China, Aug. 2025, pp. 1–6.
- **Junxiang Yang**, Kaiyuan Wang, Zhen Sun, Junming Zeng, "A Two-Stage Optimization Strategy for Enhancing the Quality Factors of Sandwich-Structured PCB Based Resonators," in *2025 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC Asia-Pacific)*, Singapore, Nov. 2025.
- Kaiyuan Wang, **Junxiang Yang**, Zhen Sun, Yun Yang, "Optimized Stacked Passive Coil Array Design for High Efficiency and Misalignment Tolerance in PCB-Based Wireless Power Transfer Systems," in *IEEE Transactions on Power Electronics*, 2026, doi: 10.1109/TPEL.2026.3680802.
- Yao Wang, **Junxiang Yang**, Yingying Song, Kaiyuan Wang, Yun Yang, "Three-Mode MHz Wireless Power Transfer: Inductive, Capacitive, and Hybrid, With a Unified Coupling Architecture," in *IEEE Journal on Wireless Power Technologies*, 2026, doi: 10.1109/JWPT.2026.3680415.
- Kaiyuan Wang, **Junxiang Yang**, Honghong Pei, Yun Yang, "Non-Uniform Resonant Frequency Design for Performance Enhancement in Wireless Power Transfer Systems with Relay Resonators," in *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, 2026, doi: 10.1109/JESTIE.2026.3706827.
- Xiaojun Deng, Noven Lee, **Junxiang Yang**, Yajie Jiang, Yun Yang, "Offset Mitigation of a Switched Capacitor Single-Input-Multioutput Converter Using an Artificial Neural Network Based Virtual Reference Control," in *IEEE Transactions on Power Electronics*, Apr. 2026, doi: 10.1109/TPEL.2025.3626401.

- Yao Wang, **Junxiang Yang**, Kaiyuan Wang, Yun Yang, "Highly Integrated Hybrid Inductive and Capacitive Power Transfer System with Asymmetrical Printed-Circuit-Board-Based Self-Resonator," in *IEEE Transactions on Power Electronics*, vol. 40, no. 7, pp. 10254–10264, July 2025.
- Kaiyuan Wang, **Junxiang Yang**, Zhen Sun, Ching-Ming Lai, Yun Yang, "Modeling of Kilohertz Wireless Power Transfer Systems with Transmitter-Side Low-Frequency Metamaterials," in *2025 IEEE Energy Conversion Congress and Exposition (ECCE)*, pp. 1–7, Oct. 2025, doi: 10.1109/ECCE58356.2025.11259972.
- Kaiyuan Wang, Jinhang Li, Zheyuan Bai, **Junxiang Yang**, Yao Wang, "A Comparative Study of Kilohertz Wireless Power Transfer Systems Using Different Transmitter-Side Low-Frequency Metamaterials," in *Annual Conference of China Electrotechnical Society*, pp. 184–193, 2025, doi: 10.1007/978-981-95-8252-5_20.
- Kaiyuan Wang, **Junxiang Yang**, Ching-Ming Lai, Yun Yang, "Equivalent Circuit Modeling and Analysis of Low-Freqeuncy Electromagnetic Metasurface-Based Wireless Power Transfer Systems," in *2025 IEEE 20th Conference on Industrial Electronics and Applications (ICIEA)*, pp. 1–5, Aug. 2025, doi: 10.1109/ICIEA65512.2025.11
- Kaiyuan Wang, **Junxiang Yang**, Zhen Sun, Junming Zeng, and Yun Yang, "Analysis and design of domino inductive power transfer system with enhanced quasi-load-independent constant current output," in *2024 IEEE Wireless Power Technology Conference and Exposition (WPTCE)*, May 2024.
- Yajie Jiang, **Junxiang Yang**, Jiayang Wu, Kerui Li, Yun Yang and S. C. Tan, "A secured fast charging control based on a machine learning-aided electrothermal model for lithium-ion batteries," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, Oct. 2024.
- Shuye Shang, **Junxiang Yang**, Tenghao Ji, Minghao Fan, Kaiyuan Wang, and Heshou Wang, "Investigations of a Simplified PCB-Based Wireless Power Resonator Operating at 13.56 MHz," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.
- Shuye Shang, Kaiyuan Wang, **Junxiang Yang**, Rui Liang, Zhen Sun, Yao Wang, Yun Yang, "Omnidirectional Wireless Charging of Battery-Free Drones in Octagonal Prism Chamber," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.
- Ke Tang, **Junxiang Yang**, Zhen Sun, Kaiyuan Wang, Junming Zeng, Heshou Wang, "Load and Mutual Detect Methods for Omnidirectional Wireless Power Transfer Systems Based on Third Harmonic Analysis," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.
- Zekai Zhao, Huajun Wang, Rui Liang, **Junxiang Yang**, Yajie Jiang, Kaiwen Chen, "Electric Vehicles Charging and Discharging Strategy Based on Differential Evolution (DE) Algorithm," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.
- Tade Hrutuja Sanju, **Junxiang Yang**, Rui Liang, Tenghao Ji, Shuye Shang, Yajie Jiang, "Parameter Identification of Electrical and Thermal Model of a Lithium Polymer Battery Using Particle Swarm Optimization," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024, doi: 10.1109/PESA62148.2024.10594791.
- Xiangrong Zhang, **Junxiang Yang**, Rui Liang, Tenghao Ji, Shuye Shang, Kaiyuan Wang, "High Frequency Inverter Design for Wireless Power Transfer System," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.
- Qinyu Li, Fazil Arhanifsyah, Zhen Sun, Rui Liang, **Junxiang Yang**, Yun Yang, "A Jerusalem Electromagnetic Compatibility Coat for a Cubic Wireless Charging Container," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.
- Chen Chen, Tenghao Ji, **Junxiang Yang**, Rui Liang, Kaiyuan Wang, Jinhong Sun, "Design and Analysis of Class Φ Inverters in the Wireless Power Transfer," in *10th International Conference on Power Electronics*

Systems and Applications (PESA), Jun. 2024.